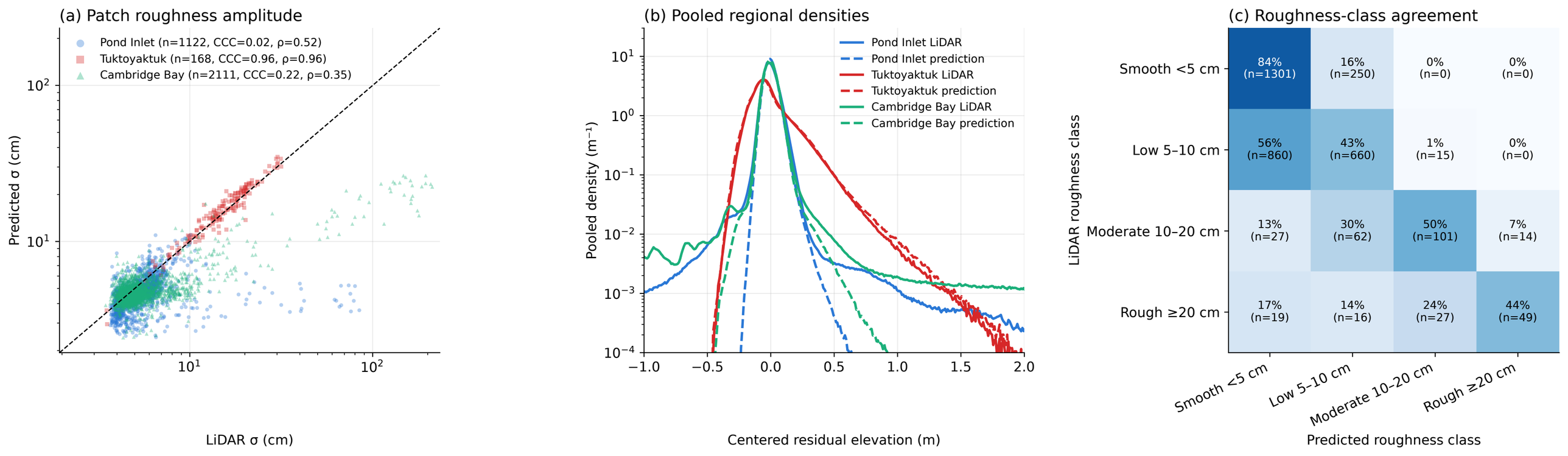

(a) Patch roughness amplitude
Pond Inlet (n=1122, CCC=0.02, ρ=0.52)
Tuktoyaktuk (n=168, CCC=0.96, ρ=0.96)
Cambridge Bay (n=2111, CCC=0.22, ρ=0.35)
Predicted σ (cm)
LiDAR σ (cm)
(b) Pooled regional densities
Pond Inlet LiDAR
Pond Inlet prediction
Tuktoyaktuk LiDAR
Tuktoyaktuk prediction
Cambridge Bay LiDAR
Cambridge Bay prediction
Pooled density (m⁻¹)
Centered residual elevation (m)
(c) Roughness-class agreement
LiDAR roughness class
Smooth <5 cm
Low 5–10 cm
Moderate 10–20 cm
Rough ≥20 cm
84% (n=1301)
16% (n=250)
0% (n=0)
0% (n=0)
56% (n=860)
43% (n=660)
1% (n=15)
0% (n=0)
13% (n=27)
30% (n=62)
50% (n=101)
7% (n=14)
17% (n=19)
14% (n=16)
24% (n=27)
44% (n=49)
Predicted roughness class